

Resilient Multimodal Geospatial Reasoning with Tool-Augmented Remote Sensing Agents

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Abstract—Beyond perception, reasoning is crucial in remote sensing, enabling advanced interpretation, inference, and decision-making. Recent advances in large language models (LLMs) have given rise to tool-augmented agents that enhance reasoning by leveraging external tools for complex analytical tasks. However, existing research on these agents in remote sensing largely focuses on perception-oriented tasks, with cognitive geospatial reasoning remaining underexplored. In this work, we systematically evaluate the geospatial reasoning capabilities of LLM-powered tool-augmented agents. To this end, we introduce GeoHIER, a benchmark for hierarchical geospatial reasoning. GeoHIER comprises 545 scenario-driven, hierarchy-aware tasks—such as hazard vulnerability assessment, urban heat island analysis, and forest fragmentation dynamics—spanning optical, Synthetic Aperture Radar (SAR), and infrared (IR) imagery. GeoHIER advances evaluation beyond monitoring-based recognition to cognitive-level geospatial analysis. Building upon GeoHIER, we propose HieraPlan, an tool-augmented agent powered by LLMs that organizes 5 toolkits into functional hierarchies and executes fault-tolerant reasoning pipelines. HieraPlan enables structured abstraction, robust recovery from tool failures, and stable long-horizon planning. Extensive experiments across diverse geospatial reasoning tasks demonstrate that HieraPlan excels in hierarchical reasoning, cross-modal transfer, and error handling. To facilitate future research, we will make our dataset and codebase publicly available at: <https://github.com/Ireliya/Resilient-Multimodal-Geospatial-Reasoning-with-Tool-Augmented-Remote-Sensing-Agents>

Index Terms—geospatial reasoning, large language models, intelligent agents

I. INTRODUCTION

Large language models (LLMs), empowered by generative pretraining and instruction tuning, have substantially improved zero-shot task completion across diverse applications [1, 2]. Building on this progress, LLM-driven agents can further decompose high-level goals into sub-tasks and orchestrate external tools, enabling robust multi-step workflows [3, 4]. Despite these advances, directly applying general-domain agents to remote sensing (RS) remains challenging: RS data exhibit strong modality heterogeneity (e.g., optical, Synthetic Aperture Radar (SAR), infrared), and targets vary drastically in scale and orientation across diverse landscapes. These characteristics often break assumptions behind generic visual-language reasoning, causing agents to degrade under RS-specific data distributions and stricter constraints on spatial units and tool parameters.

To address RS challenges, recent work has begun adapting LLMs into RS agents, such as Remote Sensing ChatGPT [5], RS-Agent [6], GeoLLM-Engine [7], Change-Agent [8], and Tree-GPT [9]. These systems demonstrate promising capability for perception-oriented tasks (e.g., classification, localiza-

tion, counting, and RS VQA). However, realistic Earth observation (EO) applications demand *cognitive-level geospatial reasoning* beyond perception, where decisions are driven by scenario intent (e.g., flood response or irrigation monitoring) and require multi-step spatial analysis that composes intermediate products (e.g., masks, polygons, buffers) into actionable outcomes. For instance, flood-risk assessment screens buildings in proximity to water bodies [10], irrigation analysis quantifies cropland adjacency to water sources [11], and ecological stabilization identifies barren patches fully enclosed by forests [12].

Recent benchmarks have begun to move from perception-oriented evaluation toward geospatial reasoning, yet they emphasize different facets and leave key gaps unresolved. ThinkGeo [13] emphasizes execution-grounded RS evaluation through multi-step trajectories, but remains narrow in task scope and modality coverage, leaving multimodal (especially infrared) settings and scenario-driven planning underexplored. Earth-Agent [14] focuses on broad integration of EO resources and large-scale tool orchestration, but provides limited treatment of standards-anchored spatial triggers, validity constraints over spatial units and tool parameters, and failure-robust execution under RS modality heterogeneity. Consequently, benchmarks and agents for cognitive-level geospatial reasoning still lack a unified formulation that is simultaneously *multimodal*, *scenario-driven*, and *tool-verifiable*.

Crucially, many EO decisions are inherently *normative*: they must satisfy standardized spatial criteria (e.g., buffer distances, ratio thresholds, and topological predicates) specified in domain guidelines. Making such criteria actionable requires an explicit *hierarchical scene context*. By hierarchical scene context, we mean a structured representation that organizes an EO problem from *scenario intent* (why we observe) to *relevant assets* (what entities matter) and further to *dependency-aware sub-goals* (how intermediate analyses compose), e.g., “delineate water extent” → “identify exposed buildings within a distance buffer” → “rank priorities by coverage ratios.” This hierarchy determines (i) which intermediate products should be produced and reused across sub-goals, (ii) which spatial units and predicates are valid at each step, and (iii) how compliance with standards-anchored criteria can be verified via tools. Yet existing benchmarks and agent frameworks do not systematically operationalize these normative criteria under such context as *runtime-checkable task specifications* with executable, verifiable tool programs. Without this anchoring, tool-executed narratives may appear plausible but remain non-reproducible, non-auditable, and difficult to compare across methods—especially under RS modality heterogeneity.

To bridge this gap, we introduce **GeoHIER**, a benchmark explicitly designed for cognitive-level geospatial reasoning across multiple RS modalities. GeoHIER comprises 545 scenario-driven tasks spanning optical, SAR, and infrared imagery. Each task is specified in a structured form that couples *standards-anchored spatial triggers* (quantitative thresholds or topological predicates) with expert interpretation and verifiable tool chains, and is organized under scenario intents with dependency-aware sub-goals to support planning and verification. Building on GeoHIER, we further propose **HieraPlan**, an LLM-driven agent that hierarchically organizes analytic tools and executes constraint-aware, fault-tolerant reasoning pipelines with recovery from intermediate failures.

Our main contributions are summarized as follows:

- We present **GeoHIER**, the first benchmark for hierarchical geospatial reasoning, comprising 545 tasks across optical, SAR, and infrared modalities.
- We propose **HieraPlan**, an agentic framework that enables cognitive reasoning with hierarchical planning, robust error handling, and support for multi-modal RS scenarios.
- We establish new baselines by comprehensively evaluating state-of-the-art LLMs on **GeoHIER**, revealing both their strengths and limitations in cognitively demanding geospatial reasoning.

II. RELATED WORK

The extension of LLMs to Earth observation has rapidly progressed from perception-oriented assistants to agentic systems that orchestrate external tools and support multi-step geospatial workflows. Early RS agents predominantly target specialized perception problems. For example, TreeGPT [9] focuses on forestry applications by integrating tree-level delineation with ecological parameter extraction, while Change-Agent [8] supports bi-temporal change detection and interactive captioning to interpret changed regions. These task-focused systems demonstrate the promise of coupling LLM interfaces with RS expert models, yet their operational scope is often bounded by short tool chains and narrow task abstractions, limiting their ability to address cognitively demanding scenarios that require explicit task hierarchies, long-horizon decomposition, and robust tool orchestration.

Beyond task-specific agents, recent efforts aim to build more general-purpose RS agents by expanding tool coverage, executable interfaces, and code-generation capabilities. Remote Sensing ChatGPT [5] integrates ChatGPT [16] with pretrained RS networks to handle a range of perception-oriented tasks. RS-Agent [6] further broadens the task spectrum through scalable tool integration, enabling workflows that require specialized domain expertise. In parallel, GeoLLM-Engine [7] provides an executable geospatial environment with fully operational APIs and interactive map/web interfaces, while UnivEARTH [15] curates domain-grounded QA tasks from NASA Earth Observatory articles to evaluate whether LLMs can generate executable Google Earth Engine code. Collectively, these systems advance executability and tool

access; however, tools are often exposed as a largely flat action space, leaving hierarchy-aware tool retrieval, structured task abstraction, and constraint-aware planning underexplored.

A closely related research thread is the emergence of *execution-grounded* evaluation and benchmarks, which move beyond perception-only accuracy by testing models in executable tool environments with multi-step workflows. Two representative directions have recently crystallized. First, ThinkGeo [13] introduces an agentic benchmark grounded in real RS imagery, providing ReAct-style annotated tool trajectories and execution-level metrics in an executable environment tailored to RS workflows. Second, Earth-Agent [14] pushes toward a broader EO agent stack by integrating a large toolkit (104 specialized tools grouped into functional kits) and proposing Earth-Bench to evaluate multi-step quantitative EO analysis over RGB imagery, raw spectral data, and processed EO products, together with trajectory-aware evaluation protocols. While these benchmarks differ in scope (RS-specific benchmarking versus integrated EO tool ecosystems), both reflect the broader shift toward execution-grounded assessment and process-aware evaluation.

Despite recent progress, most existing benchmarks define reasoning mainly as the execution of tool sequences. They do not explicitly specify scenario-level intent, dependency-aware subgoals, or standards-based spatial criteria as part of the task definition. As shown in Table I, current benchmarks usually support *verifiable tool execution* and *step-level evaluation*; however, they often represent problems as flat sequences of tool calls, without modeling hierarchical scene context or domain semantics such as spatial triggers. In addition, reasoning across heterogeneous remote-sensing modalities—optical, SAR, and infrared—is only partially evaluated. These limitations highlight the need for benchmarks that simultaneously support modality diversity, verifiable execution, and hierarchy-structured geospatial reasoning.

III. GEOHIER DATASET

We propose **GeoHIER**, a benchmark designed to assess the geospatial reasoning capabilities of tool-augmented agents powered by LLMs. GeoHIER integrates diverse imagery from optical, SAR, and infrared modalities with expert-curated knowledge and tool-augmented query pipelines, yielding 545 high-quality instances.

Each instance couples real-world imagery with structured, multi-step reasoning challenges that require both low-level perception (e.g., segmentation, detection) and high-level decision-making (e.g., urban planning, disaster assessment). Unlike prior remote sensing benchmarks that focus narrowly on perception (e.g., classification or detection), GeoHIER emphasizes *end-to-end reasoning*, from perception through spatial analysis to actionable conclusions, thereby providing a rigorous testbed for agents. Figure 1 illustrates six representative samples from the GeoHIER benchmark.

A. Geospatial Reasoning Scenarios

A central challenge in RS is bridging the gap between low-level perceptual outputs and the high-level intelligence

TABLE I: Comparison of representative **execution-grounded remote sensing (RS/EO) reasoning benchmarks**. (✓: fully supported; ×: not supported; –: not emphasized or not applicable.)

Benchmark	Multi-sensor (Optical/SAR/IR)	Verifiable execution	Step-level eval.	Hierarchical scene context
UnivEARTH [15]	×	✓	–	–
GeoLLM-Engine [7]	×	✓	✓	–
ThinkGeo[13]	×	✓	✓	–
Earth-Bench[14]	×	✓	✓	–
GeoHIER (Ours)	✓	✓	✓	✓

required for real-world decision-making. To address this, we adopt a structured taxonomy of geospatial reasoning tasks, systematically organized into seven primary domains: urban planning, disaster assessment, environmental monitoring, transportation analysis, activation monitoring, maritime monitoring, and industrial sites.

B. Data Construction Pipeline

We construct GeoHIER with a two-stage pipeline (see Figure 2) that combines diverse source imagery, knowledge-augmented scenario generation, and multi-pass expert adjudication to produce 545 validated instances. The source datasets used in this pipeline are summarized in Table II.

a) Stage 1: Knowledge-augmented scenario generation.:

We obtain a stratified sample of source imagery (by modality and scene type) and apply explicit hardness controls to select cases that require multi-tool reasoning and compositional spatial relations (e.g., proximity, containment, topology). Candidate queries and tool-chains are generated with ChatGPT-5 via in-context learning: prompts are modality-specific and seeded with expert-authored exemplars.

To ground generation in domain knowledge, we inject a compact knowledge corpus into prompts. The corpus covers four guidance categories: (i) urban greening and heat-mitigation frameworks [17, 18, 19, 20, 21]; (ii) international land-cover standards [22, 23]; (iii) aviation and maritime search-and-rescue doctrine for IR small-target tasks [24]; and (iv) industrial safety rules for separation and proximity of hazardous assets [25, 26]. These anchors force the ChatGPT-5 to produce quantitative thresholds (distances, areas, class compositions) consistent with established frameworks. We also enforce operator legality checks (including ground distance sampling (GSD)-aware units) to prevent invalid tool sequences.

b) Stage 2: Expert adjudication and quality gating.:

We perform expert-led curation with an *execution-gated* release criterion. Specifically, each Stage 1 candidate is subjected to a three-round vetting procedure by eight remote-sensing experts, aiming to produce instances that are both unambiguous in intent and verifiable through tool execution.

Checklist-based assessment. For every candidate, reviewers examine three aspects: (i) *semantic soundness* (the query states a well-posed, domain-relevant task without underspecification), (ii) *workflow validity* (the operator choices, ordering, and arguments comply with modality constraints and remain consistent under GSD-aware units), and (iii) *geospatial*

correctness (no geometry/topology pitfalls, and intermediate spatial objects are meaningful for downstream operators). We iteratively revise each candidate until all criteria are satisfied; otherwise, we discard it if the intended question cannot be substantiated by executable tool evidence (invalid units, ill-defined spatial predicates, or inconsistencies between the query and the expected tool outputs).

Execution-gated verification. To avoid releasing instances that rely on free-form generation, we require that the finalized tool sequence can be *run to completion* under operator legality constraints (typed I/O, unit validity, and geospatial constraints). We accept an instance only when (a) the tool chain executes without violations, (b) intermediate outputs pass basic validity checks (non-empty outputs where required and valid topology when applicable), and (c) any reported quantities in the final answer are directly computed from tool outputs rather than inferred without evidence. This gating criterion ensures that every released instance is paired with an executable tool chain that supports traceable, evidence-grounded evaluation.

Rounds and traceability. The three rounds consist of an initial screening, disagreement resolution for borderline cases, and a final conformance check across the query text, tool arguments, and produced outputs. All modifications to the query, operator selection, arguments, and final answer are logged in the curation interface, preserving instance-level traceability for verification and subsequent analysis.

C. Benchmark Statistics

GeoHIER contains 545 queries across optical, SAR, and IR modalities. In total, 13 tools are invoked 1649 times, with most queries requiring multi-tool composition. Query lengths range from short factual questions to complex compositional ones, demonstrating both diversity and reasoning depth. Detailed statistics are reported in Table III.

Each reasoning task in GeoHIER is instantiated as a distinct scene context type, characterized by three components: (i) a SPATIAL TRIGGER (quantitative thresholds or topological relations), (ii) an EXPERT INTERPRETATION (domain-grounded semantics), and (iii) a VERIFIABLE TOOL-CHAIN. This hierarchical design bridges perception and cognition, offering structured priors and explicit multi-step pathways while ensuring interpretability and verifiability. By covering both canonical and high-complexity reasoning scenarios, GeoHIER provides rigorous testbeds for evaluating diverse agentic capabilities, including fine-grained spatial understanding, multi-step tool

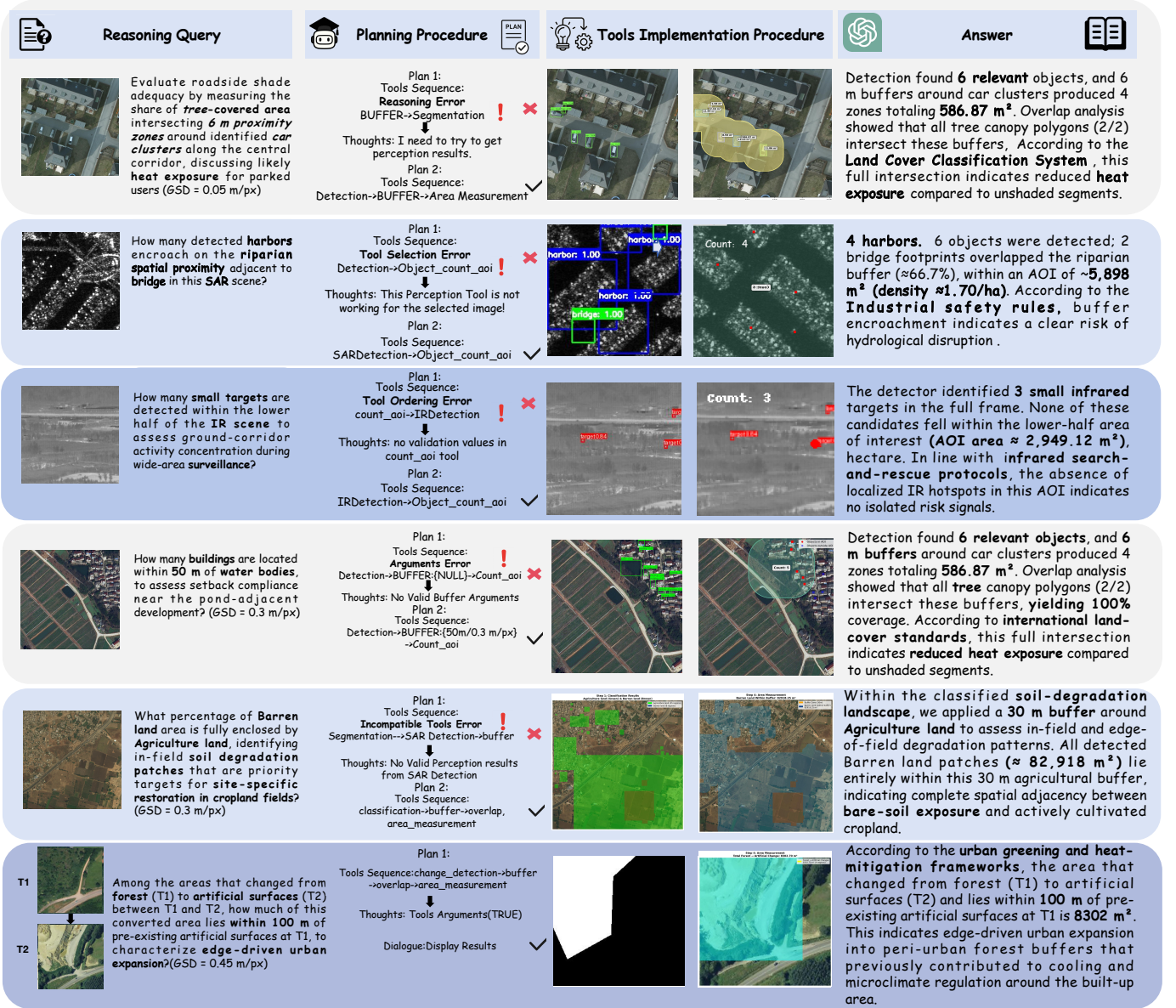


Fig. 1: Representative GeoHIER tasks and HieraPlan execution traces under our error taxonomy. From left to right: (i) input image and query, (ii) high-level plans from HieraPlan, (iii) executed tool sequence, and (iv) final geospatial answer. Red crosses mark erroneous intermediate plans or tool calls, while green checkmarks indicate the validated final tool chain and answer.

TABLE II: Datasets used as image sources in the GeoHIER benchmark.

Name	Annotation Type	Resolution	Modality	Geographical Coverage
LoveDA [27]	Masks	0.3 m/px	Optical	Nanjing/Changzhou/Wuhan, China
ISPRS Potsdam [28]	Masks	0.05 m/px	Optical	Potsdam, Germany
ISPRS Vaihingen [29]	Masks	0.09 m/px	Optical	Vaihingen, Germany
DeepGlobe Land Cover [30]	Masks	0.50 m/px	Optical	Cities in India/Indonesia/Thailand
HRSCD [31]	Change masks	0.45 m/px	Optical	Rennes/Caen, France
OGSOD [32]	Bounding boxes	3 m/px	SAR	Cities in China
DMIST [33]	Bounding boxes	—	IR	Chengdu, China

composition, quantitative analysis, and context-aware risk assessment.

IV. THE HIERAPLAN AGENT

We propose **HieraPlan**, an agent tailored for cognitively demanding geospatial reasoning in RS. HieraPlan’s LLM controller orchestrates an end-to-end workflow (see Figure 3): (i)

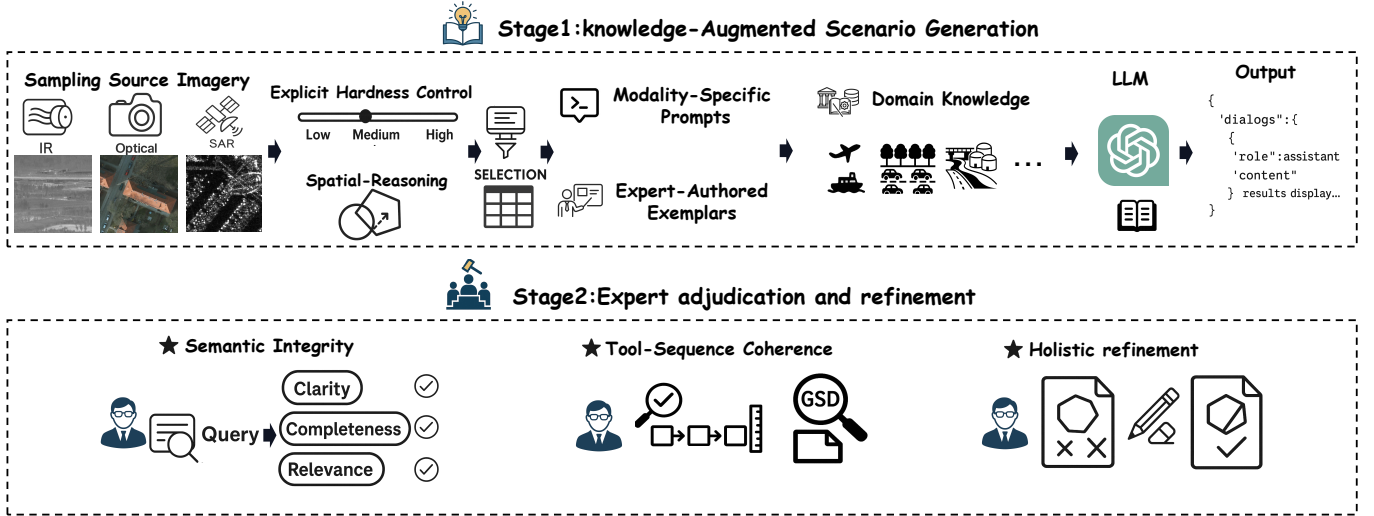


Fig. 2: Pipeline for constructing the GeoHIER benchmark.

TABLE III: Statistics of question complexity and tool-use patterns across different modalities.

Metric	All	Optical	SAR	IR
Total queries	545	386	116	43
Total tool calls	1649	1217	346	86
2/3/4-tools use	72/387/86	27/273/86	2/114/0	43/0/0
Maximum question length	N/A	406	207	288
Minimum question length	N/A	109	111	135
Average question length	158.24	160.13	143.49	181.05

semantic retrieval of a task-relevant toolset from a hierarchical, typed tool library; (ii) abstract, modality-aware plan generation over the retrieved tools; (iii) execution-time operator selection and parameterization with error-aware adaptation (within-toolkit substitution and prefix-preserving replanning); and (iv) verifiable answer synthesis strictly from structured tool outputs.

HieraPlan extends the toolkit paradigm [34] and agent framework [35] to geospatial analysis by replacing flat tool sets with a multi-level, domain-specific hierarchy that incorporates typed I/O and geospatial constraints, thereby enabling reliable, modality-aware reasoning across optical, SAR, and IR data. HieraPlan is explicitly designed to meet four RS-specific requirements: (1) *modality awareness* enforced in both retrieval and composition; (2) *hierarchical abstraction* with typed operators plus spatial unit checks and legality constraints; (3) *fault tolerance* via within-toolkit substitution and prefix-preserving replanning with structured error context; and (4) *verifiable grounding*, validating spatial outputs (e.g., geometry validity), computing all quantitative results with tools, and having the LLM synthesize interpretable, context-aware analyses by integrating validated tool I/O with pre-encoded scenario knowledge.

A. Semantic Retrieval over a Hierarchical, Typed Tool Library

We organize all tools into a multi-level hierarchy $\mathcal{H}$ with **five** top-level toolkits: (1) *Perception* (Optical segmentation, Optical detection, Optical classification, Optical change detection), (2) *Spatial-Relations* (buffer, overlap, containment),

(3) *Spatial-Statistics* (distance calculation, area measurement, object counting), (4) *SAR Tools* (SAR detection, SAR classification), and (5) *IR Tools* (IR detection). Each operator advertises (i) typed I/O (e.g., vector geometries, rasters, masks, bounding boxes), (ii) spatial unit requirements, and (iii) legality constraints, and is annotated with modality tags.

Given a natural-language query Q , a semantic retriever $\mathcal{R}$ aligns Q with textual tool descriptors in $\mathcal{H}$ using Sentence-Transformer embeddings [36] to obtain a candidate set:

$$\mathcal{C}_{\text{candidate}} = \mathcal{R}(Q, \mathcal{H}). \quad (1)$$

This step reduces the planning search space, enforces modality consistency, and prevents invalid tool compositions at the outset.

a) Integration of external expert models.: HieraPlan grounds high-level plans in verifiable computations by integrating state-of-the-art open-source models into the hierarchy: **RemoteSAM** [37] (*Perception*) for unified optical segmentation/detection/classification over 297 fine-grained categories; **SARATR-X** [38, 39] (*SAR Tools*), a foundation model pre-trained on 180K SAR samples for robust detection and classification; **DMIST/LASNet** [33] (*IR Tools*) for infrared target detection; and **Change3D** [40] (*Change Detection*), a video-modeling-based framework that unifies binary, semantic, and damage-level change reasoning through learnable perception frames. Together with Spatial-Relations and Spatial-Statistics operators, these expert tools form HieraPlan's operational backbone, ensuring reproducibility and scalability.

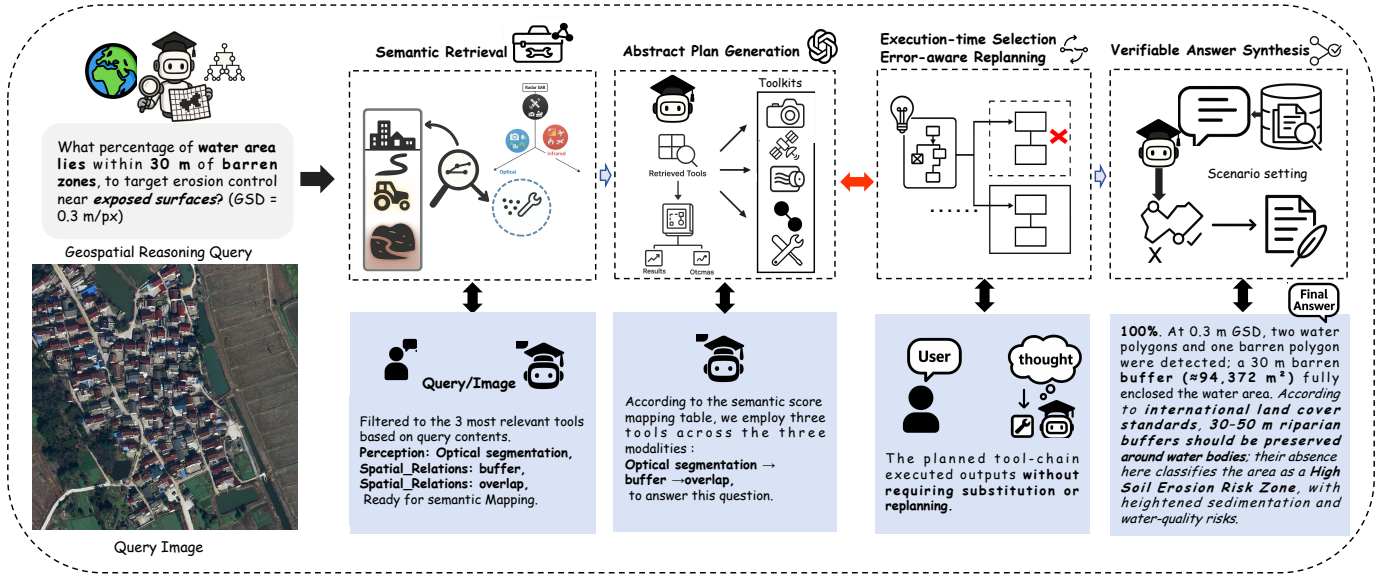


Fig. 3: End-to-end workflow of the HieraPlan Agent.

B. Abstract Plan Generation over the Retrieved Toolset

Conditioned on $\text{desc}(\mathcal{C}_{\text{candidate}})$, the planner ρ_θ (few-shot prompted with task-decomposition instructions and a constrained action schema) produces a multi-step abstract plan:

$$\mathcal{P} \sim \rho_\theta(Q, \text{desc}(\mathcal{C}_{\text{candidate}})). \quad (2)$$

Each step binds to an operator *family* in $\mathcal{C}_{\text{candidate}}$, declares expected I/O types and units, and defers parameterization (e.g., buffer distances, class labels, thresholds) to execution. This yields a modality-aware blueprint that preserves semantic validity while retaining flexibility for data-driven parameter selection.

C. Execution-Time Selection & Error-Aware Replanning

During execution, the controller instantiates each abstract step by selecting a concrete operator t , binding arguments, and validating explicit success signals (e.g., non-empty geometries/masks and numeric stability). Failures are treated as informative signals and handled via two strategies:

(1) **Within-toolkit substitution.** HieraPlan first selects a functionally similar alternative t' inside the same toolkit to preserve the abstract workflow $\mathcal{P}$ with minimal disruption (e.g., switching from *overlap* to *containment* within *Spatial_Relations*).

(2) **Prefix-preserving replanning (error-aware).** If local substitution fails, the validated prefix of $\mathcal{P}$ is retained, while a structured error context E (failed operator, arguments, logs, and validated intermediates) is injected into retrieval. A refreshed set $\mathcal{C}(Q, E)$ supports regeneration of the failing suffix. A step succeeds if any candidate succeeds ($\bigcup_{t \in \mathcal{C}(Q, E)} \text{succ}_t$); total failure occurs only when all fail ($\bigcap_{t \in \mathcal{C}(Q, E)} \text{fail}_t$). Retries are capped to bound cost, yielding a budget-conscious controller that preserves progress under adverse conditions.

D. Verifiable Answer Synthesis

HieraPlan synthesizes the final answer strictly from validated tool outputs: (i) all numeric values (counts, areas, distances, proportions) are computed by operators with explicit spatial units; (ii) aggregations include provenance (modality, operator names) and spatial unit conversions when necessary; and (iii) the LLM only provides natural-language contextualization. This separation reduces hallucination risk and improves transparency and reproducibility.

V. EXPERIMENTS AND RESULTS

To evaluate the reasoning and tool-use capabilities of **HieraPlan** under real-world RS scenarios, we conduct comprehensive evaluations on the **GeoHIER** benchmark. As the controller in HieraPlan is powered by LLMs, we evaluate a diverse suite of foundation models, spanning both API-based systems—GPT-3.5 [41], GPT-4o [42], and Gemini-2.5-Flash [43]—and a broad range of open-source LLMs. The open-source group covers both large-capacity models (DeepSeek-V3 [44], Qwen2.5-32B-Instruct [45]), LLaMA-3-70B-Instruct [46] and medium-sized systems such as Qwen2.5-7B [47], InternLM3-8B [48], LLaMA-3-8B [46], Mistral-7B [49], Yi-1.5-6B [50], and Phi-3-Mini [51]. All experiments are conducted on an NVIDIA A5000 GPU within the OpenCompass [52] evaluation platform.

A. Overall Performance

We evaluate **HieraPlan** on the **GeoHIER benchmark** under a ReAct-style [53] evaluation protocol, which includes both *step-by-step* and *end-to-end* modes. The step-by-step metrics (InstAcc, ToolAcc, ArgAcc, and SummAcc) measure instruction-following, tool selection, argument correctness, and summary generation, respectively. The end-to-end scores (P, O, and L) evaluate tool-augmented predictions within the *Perception*, *Spatial-Statistics*, and *Spatial-Relations* toolkits. In

our framework, HieraPlan’s hierarchical orchestration complements (rather than changes) the ReAct protocol by providing structured execution while preserving the same evaluation criteria.

For *final answer accuracy* (AnsAcc), we do not rely on deterministic string matching (e.g., GTA [54]), which is brittle to minor lexical variations and may incorrectly penalize semantically correct answers. This evaluation setting is particularly relevant in GeoHIER, where final responses are synthesized from *query semantics, tool inputs/outputs, and scenario knowledge* rather than matching a single canonical string. Following prior work that uses strong LLMs as evaluators (**LLM-as-a-judge**[55, 56, 57]) for scalable open-ended assessment, we compute **AnsAcc** by prompting GPT-4o-mini with a rubric-based evaluation prompt to determine whether the predicted answer (and its key arguments) are semantically consistent with the ground-truth target.

Table IV summarizes overall results on GeoHIER. GeoHIER introduces *multimodal, scenario-driven tasks* that compel agents to invoke multiple **toolkits** spanning perception, spatial analysis, SAR, and IR, thereby enabling a systematic evaluation of cognitive-level geospatial reasoning.

As shown in Table IV, **GPT-4o** achieves the best overall performance across both evaluation modes, leading on InstAcc, ArgAcc, SummAcc, and on P, O, and Ans. Among open-source large models, **Llama-3-70B-Instruct** ranks highest, with competitive ToolAcc and strong end-to-end results. For medium-scale open-source models, **Qwen2.5-7B-Instruct** stands out with the highest L score (Spatial Relations) across all evaluated models, despite weaker InstAcc and SummAcc. Across model families, **ArgAcc** remains the primary bottleneck, and lower ToolAcc is strongly associated with reduced end-to-end accuracy, highlighting the difficulty of reliable tool selection and faithful argument propagation in long-horizon geospatial workflows. Figure 4 presents example answers generated by different models for a single query.

B. Modality-wise Performance Decomposition

We further report a modality-wise breakdown to reveal whether a model’s overall score is driven by optical, SAR, or IR performance, and to diagnose which sensing domains remain the dominant bottlenecks.

As shown in Table V, GeoHIER performance exhibits strong modality-dependent variation. Frontier API models lead overall, with **GPT-4o** achieving the highest benchmark accuracy and excelling in both Optical (29.29%) and IR (100%) tasks. Yet even GPT-4o drops sharply on SAR (11.50%), underscoring that radar-domain reasoning remains challenging for current LLMs.

Open-source models show highly uneven behavior: some (e.g., **DeepSeek-V3**, **Qwen2.5-32B**) are competitive on Optical but collapse on SAR, with seven models scoring 0%. **Llama-3.1-8B** is the most robust in this domain, posting the highest SAR accuracy (26.72%), suggesting comparatively stronger resilience to speckle-dominated radar imagery.

IR performance is sharply polarized—frontier API models achieve perfect accuracy, while smaller open-source systems

remain below 45%—suggesting that IR performance is more sensitive to model capacity than Optical/SAR.

Overall, these results confirm that accuracy disparities across Optical, SAR, and IR are systematic, and that geospatial reasoning performance is constrained as much by sensing modality as by model scale and architecture.

C. Task-Oriented Diagnostic Analysis

To better understand the factors behind model successes and failures under these diverse settings, we analyze model behavior on GeoHIER from four complementary perspectives: (i) *macro-level execution dynamics* (task success/failure and reasoning depth), (ii) *capability-conditioned subsets* that disentangle semantic perception, geometric relations, quantitative statistics, and long-horizon planning, (iii) *error taxonomy* that localizes failures across format, reasoning, perception, and tool-use breakdowns, and (iv) *computational cost* that characterizes efficiency trade-offs in long-horizon tool-augmented reasoning.

a) *Macro-level execution dynamics.*: Figure 5 summarizes overall robustness and planning depth. In Subfigure 5(b), **API-based models** (e.g., **GPT-4o** and **GPT-3.5**) complete substantially more tasks with fewer failures than open-source systems, indicating stronger instruction adherence and more reliable tool execution. Subfigure 5(a) further shows that longer chains do not automatically translate into higher success: large open-source models (e.g., **Llama-3-70B-Instruct**) often generate longer trajectories (4.8–5.3 steps on average), whereas **GPT-4o** achieves a better depth–stability trade-off with about 4.2 steps per task. Overall, strong performance requires not only deeper plans but also *coherent step composition, correct argument binding, and stable tool-integrated execution*.

b) *Fine-grained capability subsets.*: To pinpoint where failures arise, Table VI stratifies GeoHIER into four subsets that progressively increase capability composition and planning horizon.

The four subsets in Table VI are defined as follows. **Semantic + Geometry (S+G)** captures tasks that require semantic perception followed by geometric relations (e.g., segmentation then buffering/overlap), excluding quantitative statistics. **Semantic + Quantitative (S+Q)** combines semantic perception with numerical measurement (e.g., detection plus distance/counting), without geometric predicates. **Semantic + Geometry + Quantitative (S+G+Q)** represents full-pipeline workflows integrating all three capabilities. **Long-Horizon (H)** includes instances with $K \geq 4$ sequential tool operations; we further report $H \cap (S+G+Q)$ to isolate the hardest long-horizon full-pipeline cases.

Two consistent trends emerge. First, *capability composition* is substantially harder than any single capability: performance drops from S+G or S+Q to S+G+Q for most models, indicating that errors compound when arguments must be propagated across heterogeneous toolkits. Second, *planning horizon* remains the dominant difficulty: on $H \cap (S+G+Q)$, even strong systems maintain moderate step-level accuracies but still collapse in final answer accuracy, suggesting that minor local

TABLE IV: **Main results of HieraPlan on the GeoHIER benchmark.** Inst., Tool., Arg., and Summ. denote step-by-step accuracies (InstAcc, ToolAcc, ArgAcc, SummAcc). P, O, and L denote end-to-end scores in the *Perception*, *Spatial Statistics*, and *Spatial Relations* toolkits. **Ans.** denotes the final answer accuracy (AnsAcc). **Bold** indicates the best overall score, and underline marks the best within the same model scale.

Model	Step-by-Step Metrics (%)				End-to-End Metrics (%)			
	Inst.	Tool.	Arg.	Summ.	P	O	L	Ans.
API-based								
GPT-4o	79.40	72.01	23.57	82.83	72.73	88.38	77.95	25.70
Gemini-2.5-Flash	59.09	43.31	15.75	47.22	41.47	40.89	41.51	12.04
GPT-3.5	59.05	57.12	21.88	59.08	61.22	61.04	53.30	24.40
Open-source (Large)								
DeepSeek-V3	65.84	66.33	23.32	60.92	65.44	66.04	57.93	21.28
Qwen2.5-32B-Instruct	62.83	58.00	17.41	58.90	44.28	67.40	61.47	19.08
Llama-3-70B-Instruct	67.89	67.47	22.66	69.91	62.72	71.62	62.54	22.75
Open-source (Medium/Small)								
Qwen2.5-7B-Instruct	77.19	74.71	23.11	65.32	68.56	73.03	82.42	22.57
InternLM3-8B-Instruct	61.26	57.72	22.59	50.28	61.44	53.70	60.70	18.90
LLaMA3-1-8B	69.46	65.99	22.27	55.60	68.42	60.34	70.82	16.70
Phi-3-Mini-4K-Instruct	55.45	41.14	17.17	10.09	43.63	7.52	51.74	3.67
Mistral-7B-Instruct-v0.2	60.03	47.05	16.73	46.06	49.75	44.05	45.23	13.21
Yi-1.5-6B-Chat	63.38	42.08	16.00	37.98	47.44	38.73	36.02	10.83

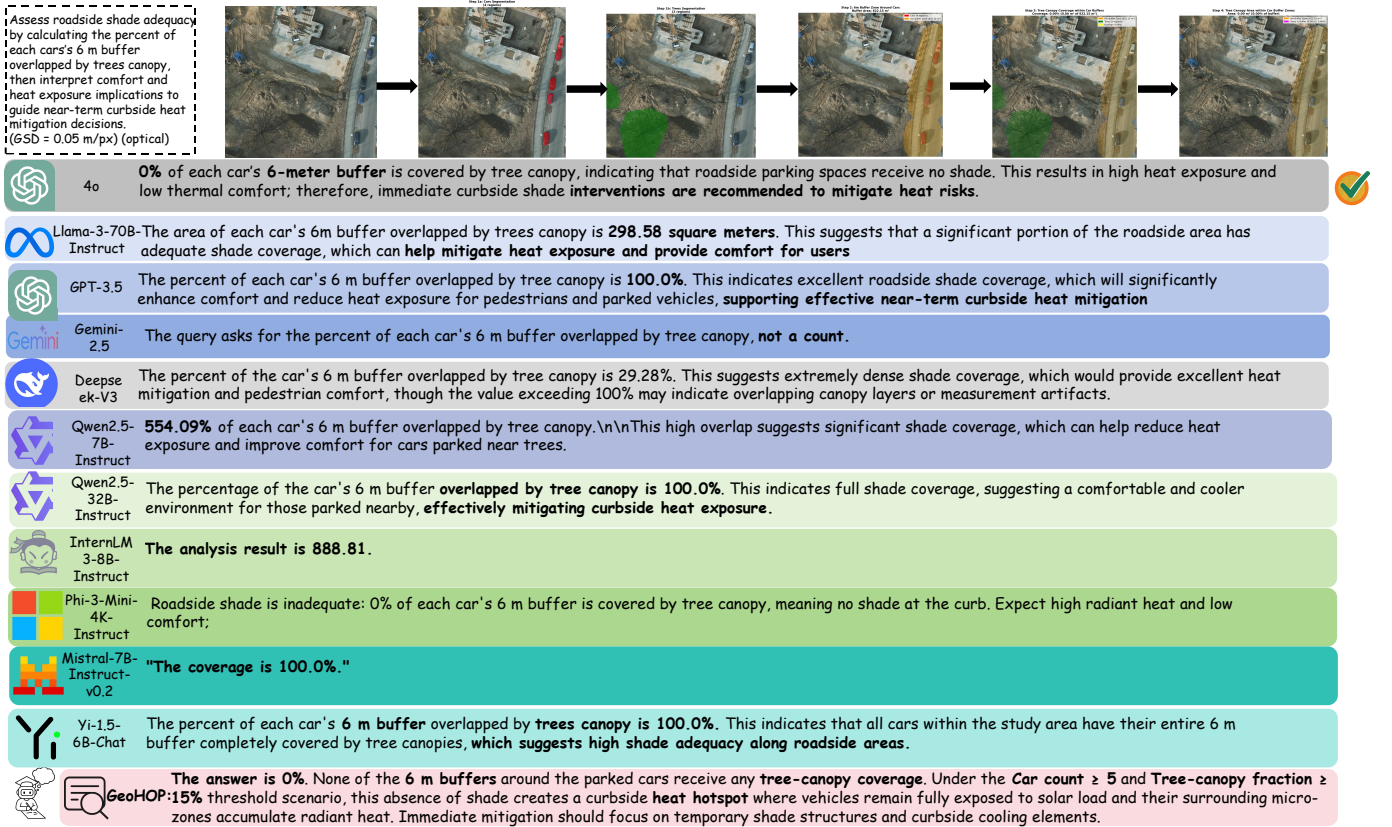


Fig. 4: Comparative evaluation of LLMs on spatial reasoning tasks.

tool/argument mistakes can derail end-to-end correctness under long chains.

c) *Task Error Analysis.*: To explain the above performance gaps, we quantify where and how failures occur via an error taxonomy, emphasizing format, reasoning, perception, and tool-use breakdowns.

As shown in Table VII, when comparing model families,

format errors emerge as the most common failure mode, particularly among open-source systems such as **DeepSeek-V3** (66.40%), **Llama-3-70B** (55.20%), and **Phi-3-Mini** (45.60%). Even frontier API models like **GPT-4o** show a 41.65% rate, indicating that producing valid, structured tool calls—especially JSON argument fields—remains the primary bottleneck in geospatial tool-augmented reasoning. Tool-use errors further

TABLE V: Accuracy comparison across models on the overall benchmark and on each sensing modality (Optical, SAR, IR).

Model	Overall (%)	Optical (%)	SAR (%)	IR (%)
GPT-4o	25.70	29.29	11.50	100.00
GPT-3.5	24.40	28.05	0.00	56.82
Gemini-2.5-Flash	12.04	11.63	0.00	100.00
DeepSeek-V3	21.28	25.19	0.00	43.18
Qwen2.5-32B-Instruct	19.08	23.12	0.00	34.09
Qwen2.5-7B-Instruct	22.57	22.08	17.24	40.91
Llama-3.1-8B-Instruct	16.70	14.81	26.72	6.82
Llama-3-70B-Instruct	22.75	27.27	0.00	43.18
InternLM3-8B-Instruct	18.90	22.08	0.86	38.64
Mistral-7B-Instruct-v0.2	13.21	16.36	0.00	20.45
Phi-3-Mini-4K-Instruct	3.67	0.52	8.62	18.18
Yi-1.5-6B-Chat	10.83	13.51	0.00	15.91

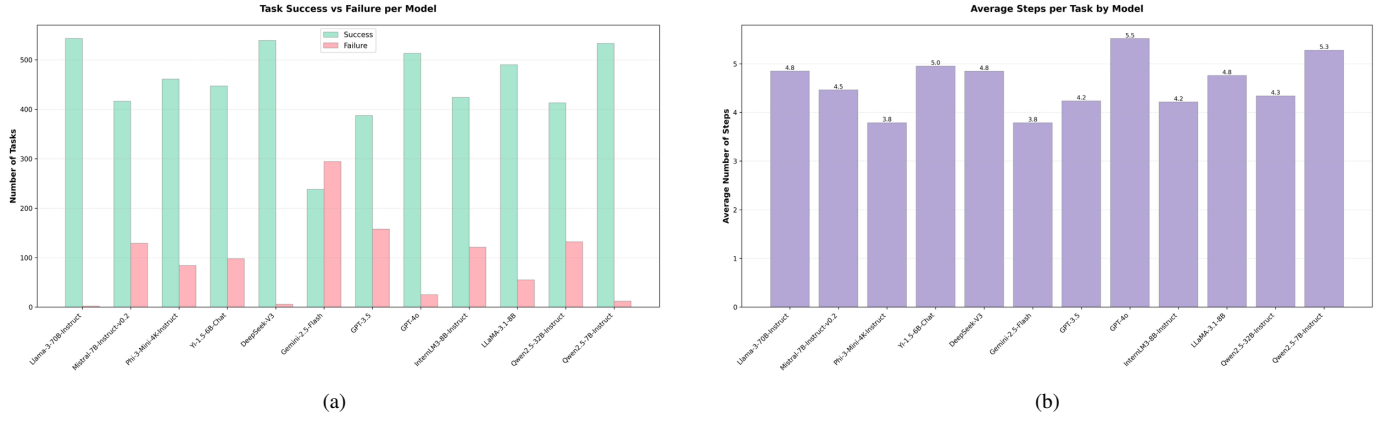


Fig. 5: Comparison of (a) task success vs. failure distribution and (b) average reasoning steps across models on GeoHIER. Frontier API-based models consistently demonstrate higher robustness and deeper tool-augmented reasoning chains, while open-source models exhibit higher failure rates and shallower execution sequences. These patterns highlight GeoHIER’s emphasis on multi-step, tool-integrated spatial reasoning and reveal structural differences in agentic planning capabilities across model families.

separate the model classes: **GPT-4o** is most reliable at 3.60%, whereas **Gemini-2.5-Flash** (37.26%) and **GPT-3.5** (39.50%) frequently misuse operators or supply invalid parameters. Low tool-use error scores for some open-source models (**DeepSeek-V3**, **Llama-3-70B**) often reflect conservative behaviour—avoiding invocation of complex tools when uncertain—rather than accurate execution. A second clear trend involves perception-driven failures, which notably impact API models. **GPT-4o** has a 41.26% perception error rate, linked to its reliance on interpreting land-cover masks, change maps, and spatial layouts. Mid-sized open-source models (e.g., **Qwen2.5-7B**, **LLaMA3-8B**) display more moderate perception errors but higher instability in reasoning or tool usage, consistent with avoiding deep perceptual grounding in favour of heuristic shortcuts. Reasoning errors are generally moderate across all systems (13–35%) and do not appear to be the main bottleneck.

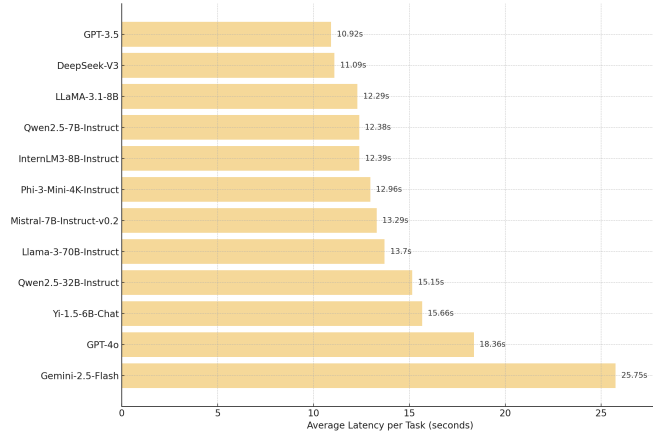
Together, these results indicate that robust structured output generation and stable multimodal perceptual grounding—not abstract reasoning—are the key limitations for current geospatial agents, highlighting the need for better format alignment and stronger cross-modal grounding in future systems.

d) Computational cost. Beyond accuracy, we examine efficiency under the full 545-task evaluation. Figures 6 and 7

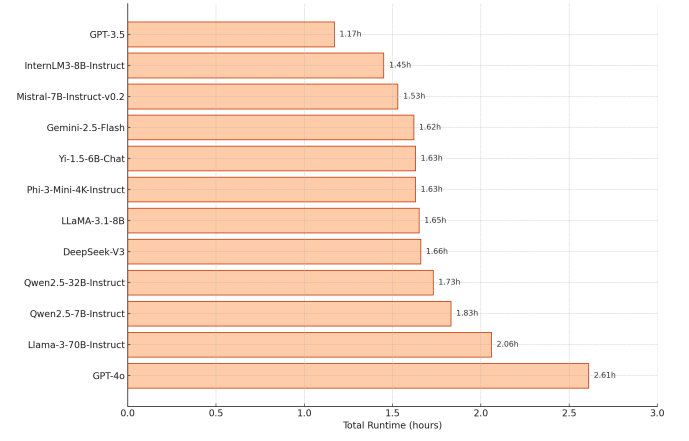
compare per-task latency, full-benchmark runtime, and the distribution of time spent on *LLM reasoning*, *perception tools*, and *spatial-analysis operators*. **GPT-3.5** and **DeepSeek-V3** are the fastest in per-task latency (10.92 s and 11.09 s), while **Gemini-2.5-Flash** (25.75s) and **GPT-4o** (18.36s) incur higher overhead, consistent with deeper multi-step reasoning. At scale, **GPT-3.5** completes all tasks in 1.17h, compared with 2.61h for **GPT-4o**, illustrating the practical trade-off between throughput and stronger planning/execution. Notably, higher-performing systems (e.g., **GPT-4o**, **Llama-3-70B**) allocate a larger fraction of execution to LLM-driven reasoning (> 55%), whereas faster but weaker models rely more heavily on tool invocations.

D. Comparison with Code-based Agent Frameworks

In addition to analyzing accuracy, failure modes, and computational cost, we further compare HieraPlan with a representative *code-based agent* paradigm to clarify complementary strengths under the same RS detection setting. Recent advances in code-based agents have shown strong potential for general-domain reasoning. Compared with traditional tool-augmented agents, these code-driven systems offer greater flexibility, richer iterative planning, and the ability to compose



(a) Per-task inference latency across evaluated models. Models with faster single-task inference exhibit lower reasoning overhead.



(b) Total runtime required to complete the full 545-task GeoHOP benchmark. Results highlight cumulative computational efficiency at scale.

Fig. 6: Comparison of model-level computational efficiency. Left: average latency per task. Right: full-benchmark total runtime.

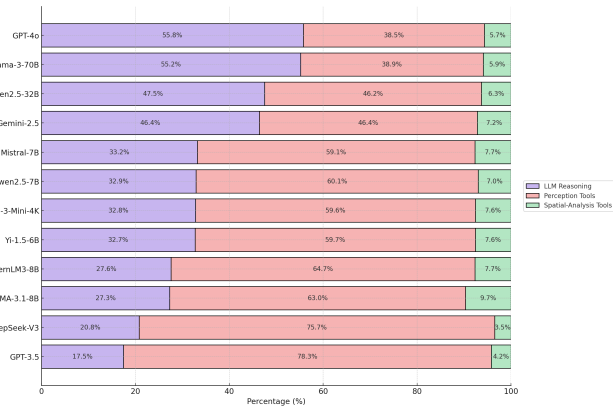


Fig. 7: Breakdown of model tool usage across the 545-task GeoHOP benchmark. Each bar shows the relative proportion of **LLM reasoning**, **perception tool calls**, and **spatial-analysis operators**. Higher-performing models tend to rely more on LLM-driven reasoning, whereas less capable models depend more heavily on perception and spatial-analysis tools.

new computational procedures—making them an increasingly influential direction in agent research.

To understand how these emerging agents behave in the RS setting, we conducted a direct comparison between a **code-based agent** (Cursor/MCP + RemoteSAM[37]) and our **tool-augmented agent** (HieraPlan) on the detection subset of GeoHIER. As shown in Table VIII, the code-based agent achieves higher final answer accuracy (44.63% vs. 40.11%), confirming its advantage in high-level reasoning. However, the tool-augmented agent retains clear strengths that are critical for real-world remote sensing. It demonstrates far superior execution stability ($O = 99.23\%$ vs. 56.50%), reflecting the reliability of pre-verified domain tools. Moreover, tool-augmented agents also provide strong controllability and avoid unsafe or hallucinated operations—factors essential in safety-sensitive RS pipelines, where algorithms such as atmospheric correction or SAR filtering must behave predictably.

These complementary properties suggest that neither paradigm is sufficient on its own. Thus, a promising direction for future RS agents is a *Hybrid Design* that integrates: (i) code-based agent-style code generation for flexible planning and adaptive reasoning, and (ii) tool-augmented agent-style verified tools for critical, high-precision RS operations.

E. Comparison with Remote Sensing Multimodal LLMs

To contextualize GeoHIER’s difficulty and position our agent against existing remote-sensing multimodal LLMs, we provide a direct comparison on the optical subset, where these RS-MLLMs are expected to be strongest. Table IX summarizes the results.

As shown in Table IX, existing **remote-sensing multimodal LLMs (RS-MLLMs)** perform poorly on the GeoHIER benchmark’s optical geospatial reasoning tasks. Models such as **EarthDial** (9.38%), **GeoChat** (6.70%), and **GeoPix** (6.17%) all struggle to exceed even 10% accuracy, despite being specifically designed for remote-sensing understanding.

In contrast, our **HieraPlan (GPT-4o)** achieves a substantially higher accuracy of **29.29%**, highlighting its stronger ability to integrate semantic cues, spatial relations, and tool-based execution. Nevertheless, the overall performance levels—both for RS-MLLMs and HieraPlan—indicate that GeoHIER’s optical tasks remain highly challenging, requiring models to handle fine-grained spatial reasoning beyond standard VQA-style interpretation.

F. Ablation Studies

To assess the contribution of HieraPlan’s core architectural components—namely the Hierarchical Tool Organization and Error-Aware Replanning—we conducted an ablation study comparing our full framework against a standard ReAct baseline using the same backbone model (GPT-4o). The ReAct baseline operates with a flat tool list (all operators provided in a single context window, without hierarchical indexing) and relies solely on a linear Thought → Action → Observation

TABLE VI: Fine-grained results across models and capability subsets. All metrics are in %.

Model	Task	Inst	Tool	Arg	Summ	Ans
API-based						
GPT-4o	S+G	77.78	92.11	32.89	75.00	45.29
	S+Q	78.33	89.20	22.80	94.00	36.00
	S+G+Q	85.20	84.40	27.37	91.08	35.43
	H $\cap$ (S+G+Q)	60.70	46.77	<u>18.09</u>	60.47	9.81
Gemini-2.5-Flash	S+G	49.44	48.03	18.42	53.57	21.43
	S+Q	30.67	25.60	12.00	30.00	8.00
	S+G+Q	34.51	23.40	9.00	24.93	9.97
	H $\cap$ (S+G+Q)	50.81	34.24	12.40	41.86	0.00
GPT-3.5	S+G	67.20	73.00	27.00	64.30	42.90
	S+Q	83.30	94.40	40.80	98.00	48.00
	S+G+Q	53.90	53.50	20.80	51.20	24.70
	H $\cap$ (S+G+Q)	67.10	<u>54.40</u>	18.30	69.80	<u>10.50</u>
Open-source (Large)						
DeepSeek-V3	S+G	80.00	86.80	28.30	92.90	32.10
	S+Q	83.30	96.80	30.00	98.00	40.00
	S+G+Q	61.90	65.70	24.00	53.80	21.30
	H $\cap$ (S+G+Q)	70.70	54.80	17.70	60.50	5.80
Qwen2.5-32B-Instruct	S+G	59.40	65.10	23.70	60.70	10.70
	S+Q	83.30	94.40	41.20	98.00	32.00
	S+G+Q	60.10	57.50	15.40	52.00	21.30
	H $\cap$ (S+G+Q)	66.00	46.50	15.20	66.30	5.80
Llama-3-70B-Instruct	S+G	80.00	80.26	24.34	89.29	25.00
	S+Q	83.33	92.80	38.80	94.00	34.00
	S+G+Q	64.44	70.12	23.17	62.99	23.10
	H $\cap$ (S+G+Q)	<u>72.21</u>	47.67	15.37	80.23	<u>11.63</u>
Open-source (Medium/Small)						
Qwen2.5-7B-Instruct	S+G	80.00	94.70	32.20	85.70	42.90
	S+Q	<u>83.30</u>	96.00	32.40	100.00	34.00
	S+G+Q	78.70	<u>77.70</u>	23.50	57.50	22.00
	H $\cap$ (S+G+Q)	69.00	53.60	17.10	73.30	12.80
InternLM3-8B-Instruct	S+G	74.40	82.90	28.90	78.60	25.00
	S+Q	83.30	94.40	41.20	100.00	32.00
	S+G+Q	55.80	55.20	22.20	38.60	19.70
	H $\cap$ (S+G+Q)	70.20	49.60	16.80	64.00	5.80
LLaMA3-1-8B	S+G	80.00	89.50	28.30	85.70	21.40
	S+Q	51.70	51.60	28.40	40.00	6.00
	S+G+Q	71.20	71.00	23.20	50.70	19.20
	H $\cap$ (S+G+Q)	67.30	48.70	16.00	<u>76.70</u>	10.50
Phi-3-Mini-4K-Instruct	S+G	51.11	42.11	14.47	32.14	7.14
	S+Q	77.67	82.00	32.40	72.00	16.00
	S+G+Q	57.35	40.98	18.34	2.62	2.62
	H $\cap$ (S+G+Q)	41.86	28.29	8.79	0.00	0.00
Mistral-7B-Instruct-v0.2	S+G	73.90	74.30	24.30	78.60	25.00
	S+Q	77.70	86.00	36.40	92.00	16.00
	S+G+Q	56.00	43.60	15.10	37.30	13.60
	H $\cap$ (S+G+Q)	65.30	40.80	14.30	47.70	5.80
Yi-1.5-6B-Chat	S+G	81.67	63.82	23.68	85.71	14.29
	S+Q	54.33	56.40	21.20	38.00	14.00
	S+G+Q	60.37	41.02	15.49	30.97	11.02
	H $\cap$ (S+G+Q)	73.37	36.82	14.60	53.49	6.98

loop, lacking HieraPlan’s type-checking, failure-recovery, and replanning modules.

Table X summarizes the results on the GeoHIER benchmark. The hierarchical tool organization in HieraGeo yields notable gains by enabling more accurate and efficient tool retrieval than the flat ReAct toolset. Even more substantial improvements appear in the end-to-end execution metrics, demonstrating the importance of our error-aware replanning module. HieraGeo proactively detects and recovers from tool-selection and argument errors, maintaining workflow continuity and significantly boosting final answer accuracy.

These ablation results confirm that HieraGeo’s structured architecture is essential for robust geospatial reasoning and

TABLE VII: Percentage distribution of four error categories for different LLMs (%). **Bold** indicates the most frequent error type for each model.

Model	Format	Reasoning	Perception	Tool-use
GPT-4o	41.65	13.49	41.26	3.60
Gemini-2.5-Flash	29.47	21.18	12.09	37.26
GPT-3.5	31.20	18.60	10.70	39.50
DeepSeek-V3	66.40	23.00	8.70	1.90
Qwen2.5-32B	31.27	19.78	13.99	34.96
Llama-3-70B	55.20	25.70	18.20	0.90
Qwen2.5-7B	42.65	23.48	29.37	4.50
InternLM3-8B	32.70	23.90	11.40	32.00
LLaMA3.1-8B	38.36	24.58	20.28	16.78
Phi-3-Mini-4K	45.60	34.70	1.80	17.90
Mistral-7B-v0.2	45.40	21.20	5.40	28.00
Yi-1.5-6B	37.90	28.60	8.60	24.90

TABLE VIII: Comparison of the Code-based Agent (CBA, i.e., Cursor) and our Tool-Augmented Agent (TAA) on the detection-task subset. $\mathcal{O}$ denotes the accuracy of spatial-statistics tool calls, $\mathcal{L}$ denotes the accuracy of spatial-relations tool calls, and **Ans.** denotes the final answer accuracy (%).

Agent Type	$\mathcal{O}$ (%)	$\mathcal{L}$ (%)	Ans. (%)
Ours (Tool-Augmented Agent)	99.23	86.51	40.11
Cursor (Code-based Agent)	56.50	85.88	44.63

TABLE IX: Optical accuracy comparison between representative remote-sensing multimodal LLMs (RS-MLLMs) and our **HieraPlan (GPT-4o)** on GeoHIER optical VQA tasks.

Model	Optical Accuracy (%)
RS-MLLMs	
EarthDial[58]	9.38
GeoChat[59]	6.70
GeoPix[60]	6.17
LLM-based Agent (Ours)	
HieraPlan (GPT-4o)	29.29

TABLE X: Ablation study comparing the full HieraGeo framework against ThinkGeo (a standard ReAct baseline) using GPT-4o. **Inst.**, **Tool.**, **Arg.**, and **Summ.** denote step-by-step accuracies, while **P**, **O**, **L**, and **Ans.** represent end-to-end execution and reasoning scores.

Method	Step-by-Step (%)				End-to-End (%)			
	Inst.	Tool.	Arg.	Summ.	P	O	L	Ans.
ThinkGeo [13] (ReAct)	77.46	48.11	14.18	46.97	48.05	5.98	50.87	8.62
HieraPlan	79.40	72.01	23.57	82.83	72.73	88.38	77.95	25.70

cannot be replicated by standard ReAct-style agents.

VI. CONCLUSION

In this work, we presented **GeoHIER**, the first benchmark explicitly designed for cognitive-level geospatial reasoning across optical, SAR, and IR modalities. GeoHIER comprises 545 hierarchy-aware, scenario-driven tasks that extend beyond perception to demand structured, multi-step reasoning. To address these challenges, we introduced **HieraPlan**, an LLM-powered agent that organizes analytic tools into functional

hierarchies, supports modality-aware planning, and ensures robust error recovery during execution. Extensive experiments on GeoHIER demonstrate that frontier models such as the GPT-4 family currently achieve the strongest performance. Simultaneously, our benchmark reveals significant potential for improvement in argument accuracy, summary generation, and multimodal toolkit integration. These findings indicate that while progress has been made, substantial headroom remains for advancing reasoning, planning, and execution capabilities in real-world remote sensing scenarios. By providing both a high-fidelity benchmark and a fault-tolerant agentic framework, our work establishes a rigorous foundation for evaluating and advancing multimodal reasoning in RS.

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